

Labelling data

The movie table is clean, but a model needs a target column, and people have to make it.

THE QUESTION

The features are collected and validated, but there is no target. How do you get labels a model can learn from?

THE INTUITION

Pick the simplest label that answers the question and write down how to decide hard cases. Have two or more people label the same items: agreement well beyond chance means the labels carry signal.

KEY TAKEAWAY

Report Cohen's kappa, not percent agreement. Aim for 0.8 or more; below 0.6, fix the guidelines first.

unlabelled rows → label shape → guidelines → pilot 50 to 100 → kappa → 2 to 3 raters + gold → adjudicate

1 The label defines the task

features: have	label: missing	
Inception 2010 \$160M	\$836M	TotalRevenue (\$) box office, studios, streaming
The Room 2003 \$6M	\$1.8M	CriticScore 0 to 100 Rotten Tomatoes, Metacritic
Parasite 2019 \$11M	\$263M	AudienceScore 0 to 100 audience and IMDb ratings
Movie #4 2015 \$45M	?	Genre tags manual annotation
... #10,000 2020 \$12M	?	

10,000 × 5 sources, converted, verified

Same rows, four targets, four problems, and each target needs its own collection. Revenue alone means about five sources per film, currency conversion and a check. ImageNet's 14M images at \$0.01 a label cost \$140K for labels alone.

2 Each task has its own label shape

Classification ☐ POS ☒ NEG ☐ NEU spam or not, sentiment "label": "NEGATIVE"	Entity spans aspirin 500 mg DRUG drug names in notes start, end, label	Bounding box  cars, people on a road [x, y, w, h]
Segmentation  tumour pixels in a scan mask[y][x] = class	Keypoints  joints for physiotherapy x, y, visibility	Audio, video  speakers, tracked objects start, end, label

Text labels can be one of N or any subset, spans, per-aspect sentiment or relations. Masks are semantic (all dogs one class) or instance (each dog its own). Tracks keep one ID across frames. Use the simplest shape that works: boxes if they suffice, not masks.

3 Spans and boxes need a written format

ORG PERSON PRODUCT LOC
 Apple CEO Tim Cook announced iPhone 15 in Cupertino.
 0 5 10 18 29 38 42 51

text[10:18] → 'Tim Cook' (end excluded)

the same "bbox": [800, 150, 100, 350]

as [x, y, w, h]

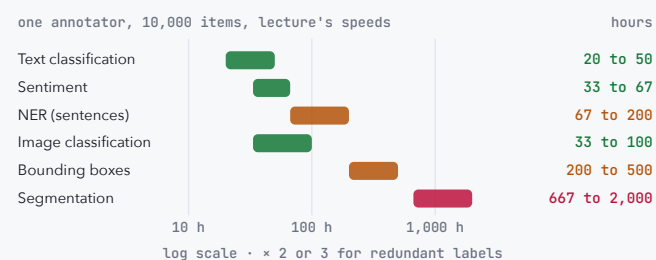
x 800 to 900, y 150 to 500

as corners [x1, y1, x2, y2]

x2 < x1: width -700

Offsets count from 0 and exclude the end, so slicing checks them. The lab's IoU code expects corners. Masks are exact but heavy, polygons compact but approximate. Keypoint flags vary; COCO uses 0 not labelled, 1 hidden, 2 visible. Span edges need rules: "New York" or "New York City"?

4 Richer labels cost 10 to 100 times more



Prices scale alike: Rs 2 against Rs 80 for a text label versus a relation (40x), Rs 2 against Rs 400 for an image label versus instance masks (200x). Experts cost more again.

5 Label Studio builds the page from XML

```
$ pip install label-studio; label-studio start → localhost:8080
```

Visually stunning but the story is predictable.

☐ Positive 1
 ☐ Negative 2
 ☐ Neutral 3
 ☒ Mixed 4

↑ <Text value="\$text"/>
 ↑ <Choices choice="single">, hotkeys 1 to 4

```
<View>
  <Text name="text" value="$text"/>
  <Choices name="sentiment" toName="text">
    <Choice value="Mixed" hotkey="4"/>
  </Choices>
</View>
```

Tasks import as [{"data": {"text": "..."}}]. choice="multiple" allows several labels, <Labels> marks spans, <RectangleLabels> draws boxes. Exports JSON, CSV, COCO, YOLO.

6 Guidelines decide the hard cases

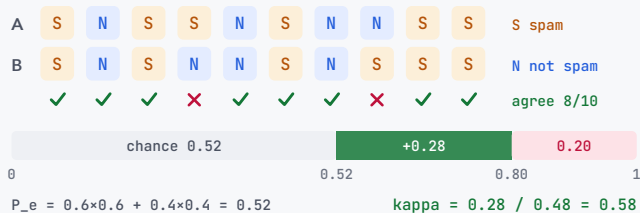
POSITIVE	Mind-blowing! Nolan does it again...	
NEGATIVE	An insult to the animated series fans.	
NEUTRAL	It's okay. Nothing special.	
MIXED	Visually stunning but the story is predictable.	
sarcasm	label the meant opinion	→ NEGATIVE
"so bad it's good"	enjoyed, but poor film	→ MIXED
still unsure	"would they recommend it?"	→ NEUTRAL
rule: "reservations → MIXED"	same page, clashing	
"Not my cup of tea but I can see the appeal."		→ NEUTRAL

Examples per label, edge cases, a rule for doubt. The lab's guideline clashes with itself (red), so raters who follow different lines will disagree.

Agreement beyond chance says the labels are usable

Kappa for categories, IoU for boxes and masks, then redundancy and a vote to clean up single slips.

7 Percent agreement counts lucky matches

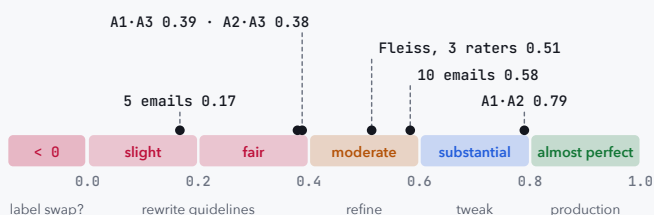


$$\kappa = \frac{P_o - P_e}{1 - P_e}, \quad P_e = \sum_c p_A(c)p_B(c)$$

Raters who each say spam 60% of the time agree 52% of the time by chance, so 80% is only 0.28 of a possible 0.48 above it. Five emails at 60% agreement give $\kappa = 0.17$.

```
from sklearn.metrics import cohen_kappa_score
cohen_kappa_score(ann_a, ann_b) # 0.58
```

8 Read kappa against fixed bands



A1 to A3 are the lab's simulated raters, right on 18, 17 and 13 of 20 reviews; the weak one pulls both its pairs to 0.38. Typical good values: 0.8 for text classes, 0.7 sentiment, 0.6 emotion.

9 Fleiss' kappa takes three or more raters

item	A1	A2	A3	Pos	Neg	Neu	Mix	P _i
1	Pos	Pos	Neu	2	0	1	0	0.33
2	Neu	Mix	Mix	0	0	1	2	0.33
3	Pos	Pos	Pos	3	0	0	0	1.00
4	Neg	Neg	Neg	0	3	0	0	1.00
5	Pos	Pos	Pos	3	0	0	0	1.00
6	Mix	Mix	Mix	0	0	0	3	1.00

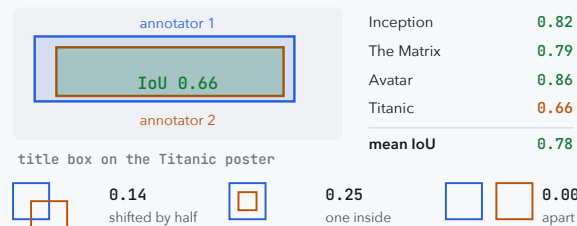
... 20 items, each row sums to 3 raters

mean P_i = 0.65 P_e = sum p_j² = 0.28 $\kappa = 0.51$

Count how many raters chose each category per item. P_i is the share of rater pairs that agree on item i ; P_e uses the pooled label shares. The result, 0.51, is near the mean of the three pairwise Cohen's kappas (0.52).

```
from statsmodels.stats.inter_rater \
import fleiss_kappa
fleiss_kappa(counts) # items x categories
```

10 Boxes, spans and transcripts use overlap



IoU is overlap over union. Above 0.5 counts as a match, 0.7 is good, 0.9 excellent; a box shifted by half its width scores only 0.14. Masks use pixel IoU or Dice = $2|A \cap B| / (|A| + |B|)$. Spans use span F1: "iPhone" against "new iPhone" misses on exact match. Transcripts use WER = $(S + I + D)/N$: "sat" to "sit" and "the" to "a" in six words is 33%. Unlike kappa, none of these subtract chance.

11 A vote over three raters fixes slips

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	
A1	P	U	P	N	P	M	P	U	P	N	P	M	U	N	M	N	P	N	P	U	18/20
A2	P	M	P	N	P	M	P	U	P	N	P	M	U	P	M	N	P	N	P	M	17/20
A3	U	M	P	N	P	M	P	U	P	M	U	P	M	U	N	M	N	P	P	P	13/20
vote	P	M	P	N	P	M	P	U	P	N	P	M	U	N	M	N	P	N	P	U	19/20
truth	P	M	P	N	P	M	P	U	P	N	P	M	U	N	M	N	P	N	P	U	

P positive · N negative · U neutral · M mixed
item 20: three different labels, no majority

Each rater alone gets 2 to 7 of 20 wrong; the majority gets 1. The lab's majority_vote uses Counter.most_common, which on a three-way tie quietly returns the first rater's label: send those items to an expert. Add about 10% gold items with known answers, calibrate until kappa passes 0.8, and track agreement over time. The lab also weights votes by each rater's accuracy.

12 Pause and think

Kappa is negative on a spam task. Why? Agreement is below chance; one rater or expert probably has the labels swapped.

IoU is 0 for boxes that plainly overlap. One tool exported [x, y, w, h] and the code reads corners. Convert first.

Kappa is 0.45 after 2,000 labels. Keep going? No. Read the disagreements, fix the guidelines, re-run a pilot. Noisy labels teach the model the noise.

Drop the weakest rater. What is left? A1 and A2 agree at 0.79, just under the 0.8 target, but two raters cannot outvote each other on a tie.

simplest label shape · written edge cases · kappa, not percent agreement · 2 to 3 raters and gold items

Source: lecture "Data labeling and annotation" and its lab (Label Studio configs, simulated raters with seed 42, IoU exercises), nipunbatra.github.io/stt-ai-teaching. The movie table comes from the data validation sheet. Getting labels with fewer people (active learning, weak supervision, LLM labels) is on the labelling-less sheet.